

Slip No. 01

Exam Seat No. _____

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

P. D. PATEL INSTITUTE OF APPLIED SCIENCE

DEPARTMENT OF NANO SCIENCE AND TECHNOLOGY

M.Sc. (1st Semester) Examination, 2009

Day: Friday

Date: 04-12-2009

Time: 10:00am-01:00pm

Sub: Computational Techniques and Programming

Sub Code: NT704

Signature of the junior supervisor and date

Notes:-

- (1) This question paper contains two parts **PART-I** and **PART-II**
- (2) **PART-I** carries **20** and **PART-II** carries **50** marks
- (3) **PART-II** contains two sections **SECTION-A** and **SECTION-B**
- (4) **SECTION-A** carries **20** and **SECTION-B** carries **30** marks
- (5) All the questions in both parts are compulsory
- (6) Figures to the right indicate the distribution of full marks for the respective question
- (7) Answers to the questions of **PART-I** should be written in the question paper itself, and, the completed **PART-I** should be submitted within the **first 20 minutes** of the examination
- (8) **PART-II** will be given after the submission of **PART-I**
- (9) Answers to the questions of this part should be written in the separate answer book

Marks of PART-I	Question Number	1 (6)	2 (14)	Total (20)
	Marks Obtained			

Signature of the Examiner

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PART-I

Total Marks: 20

1. Do as directed in the following

[2× 7=14]

(i). Find the power set of the set $\{x \mid x \text{ is a letter of the word ACHIEVEMENT}\}$

(ii). If $f(x)$ is differentiable at $x=a$, then show that $f(x)$ is continuous at $x=a$.

(iii). To show, $\int \cos x \, dx = \sin x + c, \forall x \in R$.

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(iv). Using Sandwich theorem show that $\lim_{x \rightarrow 0} \sin x = 0$.

(v). Find the cofactor of 3 in $\begin{vmatrix} 1 & 5 & 7 \\ 2 & 6 & 3 \\ 4 & 8 & 9 \end{vmatrix}$.

(vi). Find the focus and directrix of the parabola $x^2 = 8y$.

(vii). Find median of the data 23, 19, 21, 17, 22, 24, 16.

2. Out of the four alternatives A, B, C and D provided with the following questions, put the appropriate choice in the square provided there.

[1×6=6]

(i). Derivative of constant is:

- (A) 1 (B) 0 (C) constant (D) None of (A), (B), (C).

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(ii). If the relation R from a set A to a set B is equal to $(A \times B)$ then R is said to be the relation :

- (A). One One (B). Onto (C). Many One (D). Universal

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(iii). If the left and right hand limits at a point exist, are equal, but the function is not defined at this point. This type of discontinuity is:

- (A). Jump (B). Removable (C). Hole (D). Essential

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(iv). For a square matrix A, if $AA' = I$ then A is:

- (A). Symmetric (B). Orthogonal (C). Skew Symmetric (D) identity

☐

(v). Centre and radius(=r) of the circle $x^2 + y^2 - 2y = 3$

- (A). C(0,1), r=2 (B). C(0,-1), r=2 (C) C(1,0), r=4, (D). None of (A), (B), (C)

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(vi). $\frac{d}{dx} \log x$ is :

- (A). $\frac{1}{x}$ (B). $|x|$ (C) 1 (D) None of (A), (B), (C).

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Exam Seat No. _____

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M.Sc. (1st Semester) Examination, 2009

Day: Friday

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Sub: Computational Techniques and Programming

Sub Code: NT704

PART-II

Total Marks: 50

Notes:

1. Attempt all questions
2. Figures to the right indicate the distribution of full marks for the respective question
3. Answer to the question of this part should be written in the separate answer book

SECTION-A

3. Attempt the following (any two):

[4 × 2 = 8]

(a) Let $f(x) = (x-3)(9-x)$ for $3 \leq x \leq 9$,

- (i) Find $f(5)$ and $f(-2)$.
- (ii) What is the domain for definition of $f(x)$?
- (iii) Find $f(1-3t)$ and give the domain of definition.
- (iv) Find $f(f(4))$ and $f(f(6))$.
- (v) Draw graph of $f(x)$.

(b) State De Moivre's theorem.

Also, Evaluate: (i) $(-1-i)^{1/2}$ (ii) $(1-\sqrt{3}i)^5$.

(c) If $A = \{1, 2, 3, 5, 7\}$, $B = \{2, 3, 4, 6, 8\}$, $C = \{3, 4, 5, 9\}$ then verify

- (i) $(A \times B) \cup (A \times C) = A \times (B \cup C)$.
- (ii) $(A \times B) \cap (A \times C) = A \times (B \cap C)$.

4. Attempt the following (any three):

[4 × 3 = 12]

- (a) (i) Using the definition of the limit show that $\lim_{x \rightarrow 3} 6x - 9 = 9$.
 (ii) Using Sandwich theorem show that $\lim_{x \rightarrow 0} x^2 \cos(1/x) = 0$.

$$(b) f(x) = \begin{cases} 3ax + b & , x > 1 \\ 11 & , x = 1 \\ 5ax - 2b & , x < 1 \end{cases}$$

f is continuous at $x = 1$. Find a, b .

- (c) Find the area of the region bounded by the circle $x^2 + y^2 = a^2$.

(d) Evaluate : (i) $\int \frac{dx}{3 \cos x - 4 \sin x + 5}$.

(ii) $\int \sin^8 x \cos^3 x dx$.

SECTION-B**5. Attempt the following (any two):**

[4 × 2 = 8]

(a) Evaluate : $\int \frac{x^2 + 1}{x^3 - x} dx$.

(b) If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$ then prove that $A^2 - 5A - 7I = 0$.

- (c) Find the missing frequency of the following frequency distribution if the median is 37.5.

class	0-10	10-20	20-30	30-40	40-50	50-60	60-70
frequency	3	?	8	12	10	7	5

6. Attempt the following (any three):

[4×3= 12]

(a) Solve : $\sqrt{\frac{x}{x-1}} + \sqrt{\frac{1-x}{x}} = \frac{13}{6}, x \neq 0, 1.$

- (b) Calculate the mean and standard deviation for the following table giving the age distribution of 542 members.

Age in years	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No. of members	3	61	132	153	140	51	2

(c) Find the rank of $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$.

(d) (i) Find the value of k, which satisfy the equation $\begin{vmatrix} k & 1 & 0 \\ 1 & k & 1 \\ 0 & 1 & k \end{vmatrix} = 0.$

(ii) Find radius and centre of the given circle $2x^2 + 2y^2 - 28x + 12y + 114 = 0.$

7. Attempt the following (any two):

[5×2= 10]

- (a) Solve the following system of linear equations using matrix inversion

$$x - 2y + 3z = 4, 2x + y - 3z = 5, -x + y + 2z = 3.$$

- (b) Following data show particle size distribution of liquid paraffin Emulsion.

Particle size(microns)	1-30	31-60	61-90	91-120	121-150	151-180
frequency	8	13	25	29	18	7

Calculate mean, median and mode for the data.

- (c) Find the equation to the sphere through the points (0, 0, 1), (0, 1, -1), (-1, 2, 0), (1, 2, 3).